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Fri, 8 May, 2026 at 14:13

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The following submission has been created.

Track Name: Artificial Intelligence and Data Science

Paper ID: 1664

Paper Title: HierRAG: Hierarchical Retrieval-Augmented Generation with Grounded Fallback and Attribution-Aware Evaluation

Abstract:
Retrieval-Augmented Generation (RAG) systems tether language model (LM) outputs to information stored in external documents by retrieving these documents at query time. Prior systems mix three axes of evaluation: retrieval relevance, answer groundedness, and chunk influence. To date, no system has measured retrieval relevance, answer groundedness, and chunk influence simultaneously. We introduce HierRAG, our vectorless hierarchical RAG system designed to measure each axis via three interlocking mechanisms: navigational guidance via a large language model (LLM) that traverses document trees without embeddings, a groundedness-triggered retrieval cascade that retries retrieval when answers cannot be verified, and leave-one-out sensitivity testing to approximate the influence of retrieved chunks. Across 209 question-answer pairs from QASPER, HierRAG obtains an average F1 score of 0.340 overall and 0.390 on answerable questions with Gemini 3.1 Flash-Lite, while also attaining an average groundedness score of 0.825. Additionally, we found that 12.9% of questions from QASPER are unanswerable by any text-based RAG system due to structural limitations, and that retrieved chunks had absolutely no influence on the final answer text 29.7% of the time. We posit that retrieval relevance, answer groundedness, and chunk influence are three separate phenomena which require concurrent measurement.

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[hierRAG_research_2210990233_2210990221 \(1\).docx](#) (3 Mb, Fri, 08 May 2026 08:39:59 GMT)

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